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06  
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Wavelets for 2D  $\rightarrow$  Image, ...  
 $\hookrightarrow$  can be time anymore it should be discrete space, but in the literature it's usually called 2D-DWT (still DWT but for 2D) and 2D-DWPT  
non

### - - - Two-dimensional Wavelet Transform - - -

- ▶ Let  $s[\cdot][\cdot]$  be an  $N \times M$  discrete-space image, represented as a matrix, where both  $N$  and  $M$  are powers of two. Then, its 2D-DWT is another  $N \times M$  matrix obtained as follows:

- ▶ STEP 1: replace each row in  $s[\cdot][\cdot]$ , considered separately as an  $M$ -sample long signal, by its corresponding DTWT, creating the modified matrix  $\tilde{s}[\cdot][\cdot]$ ;
- ▶ STEP 2: replace each column in  $\tilde{s}[\cdot][\cdot]$ , considered separately as an  $N$ -sample long signal, by its corresponding DTWT, creating the 2D-DWT.

- ▶ All the families and filters we have studied for one-dimensional signals (1D), can be used with 2D signals. Accordingly, all the rules, tips and tricks, including Mallat's algorithm and its exceptions, are also valid for 2D signals.
- ▶ Frequently, we have  $N = M$ , i.e., the input is a square matrix.

- ▶ **Numerical Example:** Letting  $s[\cdot][\cdot] = \begin{pmatrix} 4 & 6 & 10 & 12 \\ 2 & 6 & 8 & 12 \\ 1 & 4 & 6 & 7 \\ 0 & 3 & 2 & 1 \end{pmatrix}$  and using Haar filters, let's calculate  $2D-DWT_1(s[\cdot][\cdot])$ , i.e., the first-level 2D-DWT.

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matrix  $\tilde{s}[\cdot][\cdot]$

generated from step 1

very  
book  
time on  
SE-7  
1st DWT

job  
the  
non  
1st  
column  
in the  
intermediate

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► **Solution:** From STEP 1, we get the following:

- DTWT({4, 6, 10, 12}) =  $\{\frac{10}{\sqrt{2}}, \frac{22}{\sqrt{2}}, \frac{-2}{\sqrt{2}}, \frac{-2}{\sqrt{2}}\};$
- DTWT({2, 6, 8, 12}) =  $\{\frac{8}{\sqrt{2}}, \frac{20}{\sqrt{2}}, \frac{-4}{\sqrt{2}}, \frac{-4}{\sqrt{2}}\};$
- DTWT({1, 4, 6, 7}) =  $\{\frac{5}{\sqrt{2}}, \frac{13}{\sqrt{2}}, \frac{-3}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\};$
- DTWT({0, 3, 2, 1}) =  $\{\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}, \frac{-3}{\sqrt{2}}, \frac{1}{\sqrt{2}}\}.$

Thus, the intermediary matrix needed is  $\bar{s}[\cdot][\cdot] = \begin{pmatrix} \frac{10}{\sqrt{2}} & \frac{22}{\sqrt{2}} & \frac{-2}{\sqrt{2}} & \frac{-2}{\sqrt{2}} \\ \frac{8}{\sqrt{2}} & \frac{20}{\sqrt{2}} & \frac{-4}{\sqrt{2}} & \frac{-4}{\sqrt{2}} \\ \frac{5}{\sqrt{2}} & \frac{13}{\sqrt{2}} & \frac{-3}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{3}{\sqrt{2}} & \frac{3}{\sqrt{2}} & \frac{-3}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$ , where

re the numbers in **red** and **green** are the corresponding **trends** and **fluctuations**, respectively, for each row.

► From STEP 2 and using  $\bar{s}[\cdot][\cdot]$  as input, we get the following:

- DTWT( $\{\frac{10}{\sqrt{2}}, \frac{8}{\sqrt{2}}, \frac{5}{\sqrt{2}}, \frac{3}{\sqrt{2}}\}$ ) =  $\{9, 4, 1, 1\};$
- DTWT( $\{\frac{22}{\sqrt{2}}, \frac{20}{\sqrt{2}}, \frac{13}{\sqrt{2}}, \frac{3}{\sqrt{2}}\}$ ) =  $\{21, 8, 1, 5\};$
- DTWT( $\{\frac{-2}{\sqrt{2}}, \frac{-4}{\sqrt{2}}, \frac{-3}{\sqrt{2}}, \frac{-1}{\sqrt{2}}\}$ ) =  $\{-3, -3, 1, 0\};$
- DTWT( $\{\frac{-2}{\sqrt{2}}, \frac{-4}{\sqrt{2}}, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\}$ ) =  $\{-3, 0, 1, -1\}.$

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► Consequently,  $2D-DWT_1(s[\cdot][\cdot]) = \begin{pmatrix} 9 & 21 & -3 & -3 \\ 4 & 8 & -3 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 5 & 0 & -1 \end{pmatrix}$ , where the numbers in **magenta** and **teal** are the corresponding **trends** and **fluctuations**, respectively, for each column.

► Notably, for any  $N \times M$  input matrix  $s[\cdot][\cdot]$ , its  $2D-DWT_1$  corresponds to grouping together **four**  $\frac{N}{2} \times \frac{M}{2}$  matrices, using the following disposition:

*the middle & the right &*  $\rightarrow \left( \begin{array}{c|c} \left( \begin{array}{c} \frac{N}{2} \times \frac{M}{2} \\ \text{matrix } a[\cdot][\cdot] \end{array} \right) & \left( \begin{array}{c} \frac{N}{2} \times \frac{M}{2} \\ \text{matrix } b[\cdot][\cdot] \end{array} \right) \\ \hline \left( \begin{array}{c} \frac{N}{2} \times \frac{M}{2} \\ \text{matrix } c[\cdot][\cdot] \end{array} \right) & \left( \begin{array}{c} \frac{N}{2} \times \frac{M}{2} \\ \text{matrix } d[\cdot][\cdot] \end{array} \right) \end{array} \right), \text{ where:}$

- $a[\cdot][\cdot]$  results from trending the rows of  $s[\cdot][\cdot]$  and subsequently trending the columns of  $\bar{s}[\cdot][\cdot]$ ;
- $b[\cdot][\cdot]$  results from fluctuating the rows of  $s[\cdot][\cdot]$  and subsequently trending the columns of  $\bar{s}[\cdot][\cdot]$ ;
- $c[\cdot][\cdot]$  results from trending the rows of  $s[\cdot][\cdot]$  and subsequently fluctuating the columns of  $\bar{s}[\cdot][\cdot]$ ;
- $d[\cdot][\cdot]$  results from fluctuating the rows of  $s[\cdot][\cdot]$  and subsequently fluctuating the columns of  $\bar{s}[\cdot][\cdot]$ ,

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$a[\cdot][\cdot] \rightarrow h[\cdot] \quad l[\cdot]$   
 $b[\cdot][\cdot] \rightarrow g[\cdot] \quad h[\cdot]$   
 $c[\cdot][\cdot] \rightarrow h[\cdot] \quad g[\cdot]$   
 $d[\cdot][\cdot] \rightarrow g[\cdot] \quad g[\cdot]$

*filter used each part*

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where, as we know, trend and fluctuation correspond to the content of the left and the right leaves, respectively, in the decomposition trees.

- By definition, to proceed to the second level 2D-DWT, i.e., 2D-DWT<sub>2</sub>(s[·][·]), we simply adopt the same procedure just presented **but** using **only** the small  $\frac{N}{2} \times \frac{M}{2}$  matrix a[·][·] as input, letting b[·][·], c[·][·] and d[·][·] intact. Thus, if both N and M permit, then the decomposition can be extended to the second, third, ..., and so on, level. Just after each level, the input matrix to the next level halves its number of rows and columns.
- **Numerical Example:** Let's move forward from the previous example to calculate 2D-DWT<sub>2</sub>(s[·][·]), i.e., the second-level transformation.
- **Solution:** Initially, we look at the first-level transformation, i.e.,

$$\left( \begin{array}{c|c} \left( \begin{array}{cc} \frac{N}{2} \times \frac{M}{2} \\ \text{matrix } a[\cdot][\cdot] \end{array} \right) & \left( \begin{array}{cc} \frac{N}{2} \times \frac{M}{2} \\ \text{matrix } b[\cdot][\cdot] \end{array} \right) \\ \hline \left( \begin{array}{cc} \frac{N}{2} \times \frac{M}{2} \\ \text{matrix } c[\cdot][\cdot] \end{array} \right) & \left( \begin{array}{cc} \frac{N}{2} \times \frac{M}{2} \\ \text{matrix } d[\cdot][\cdot] \end{array} \right) \end{array} \right) = \left( \begin{array}{cc|cc} 9 & 21 & -3 & -3 \\ 4 & 8 & -3 & 0 \\ \hline 1 & 5 & 1 & -1 \end{array} \right).$$

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for next levels of DTWT you use a[·][·]



## Wavelet Transform and its Applications

### - - - Two-dimensional Wavelet Transform - - -

- Now, we use only a[·][·] =  $\begin{pmatrix} 9 & 21 \\ 4 & 8 \end{pmatrix}$  as input. From STEP 1 on page 180, we get:
  - DTWT((9, 21)) =  $\left\{ \frac{30}{\sqrt{2}}, \frac{-12}{\sqrt{2}} \right\}$ ;
  - DTWT((4, 8)) =  $\left\{ \frac{12}{\sqrt{2}}, \frac{-4}{\sqrt{2}} \right\}$ .
 Thus, the intermediary matrix needed is  $\bar{a}[\cdot][\cdot] = \begin{pmatrix} \frac{30}{\sqrt{2}} & \frac{-12}{\sqrt{2}} \\ \frac{12}{\sqrt{2}} & \frac{-4}{\sqrt{2}} \end{pmatrix}$ , where the numbers in red and green are the corresponding trends and fluctuations, respectively, for each row.
- From STEP 2 on page 180 and using  $\bar{a}[\cdot][\cdot]$  as input, we get:
  - DTWT( $\left( \frac{30}{\sqrt{2}}, \frac{12}{\sqrt{2}} \right)$ ) = {21, 9};
  - DTWT( $\left( \frac{-12}{\sqrt{2}}, \frac{-4}{\sqrt{2}} \right)$ ) = {-8, -4}.
- Consequently, 2D-DWT<sub>2</sub>(s[·][·]) =  $\begin{pmatrix} 21 & -8 & -3 & -3 \\ 9 & -4 & -3 & 0 \\ \hline 1 & 5 & 1 & -1 \end{pmatrix}$ , where the numbers in magenta and teal are the corresponding trends and fluctuations, respectively, for each column. In addition, the numbers in black correspond just to the small matrices b[·][·], c[·][·] and d[·][·], from the first level.

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the new matrix is sub-matrix of a[·][·].  
In this case we can't continue the decomposition just each sub-matrix has size = 1

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- ▶ Notably, the small matrix  $\begin{pmatrix} 21 & -8 \\ 9 & -4 \end{pmatrix}$  from the previous page should also be interpreted as four smaller matrices grouped together:

- ▶  $a[\cdot][\cdot]$  in the second level, i.e.,  $a_2[\cdot][\cdot] = \begin{pmatrix} 21 \\ 9 \end{pmatrix}$ ,
- ▶  $b[\cdot][\cdot]$  in the second level, i.e.,  $b_2[\cdot][\cdot] = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$ ,
- ▶  $c[\cdot][\cdot]$  in the second level, i.e.,  $c_2[\cdot][\cdot] = \begin{pmatrix} 9 \\ -4 \end{pmatrix}$ ,
- ▶  $d[\cdot][\cdot]$  in the second level, i.e.,  $d_2[\cdot][\cdot] = \begin{pmatrix} -8 \\ -4 \end{pmatrix}$ .

Additionally, we note that since those four small matrices have only one element each, there is no possibility to proceed to the third level.

Consequently, the second-level is the last one in this case.

- ▶ **REMARK:** 2D-DWTs, at any level, preserve the input matrix energy! For usual images, most part of the energy in the transformed signal is in matrix  $a[\cdot][\cdot]$  at the deepest level. Thus, as in the 1D case, this is relevant for compression!

*The energy is preserved*

$$\underbrace{m[\cdot][\cdot]}_{\substack{Y_0 \\ Y_1 \\ Y_2 \\ Y_3}} \quad S'[\cdot][\cdot] \quad \underbrace{X[\cdot][\cdot]}_{\substack{Y_0 \\ Y_1 \\ Y_2 \\ Y_3}}$$

Now, multiplying  $m[\cdot][\cdot]$  by the transpose of  $X[\cdot][\cdot]$ , we have:

$$\underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}}_{m[\cdot][\cdot]} \cdot \underbrace{\begin{pmatrix} \frac{10}{\sqrt{2}} & \frac{-2}{\sqrt{2}} & \frac{22}{\sqrt{2}} & \frac{-2}{\sqrt{2}} \\ \frac{8}{\sqrt{2}} & \frac{-4}{\sqrt{2}} & \frac{20}{\sqrt{2}} & \frac{-4}{\sqrt{2}} \\ \frac{5}{\sqrt{2}} & \frac{-3}{\sqrt{2}} & \frac{13}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{3}{\sqrt{2}} & \frac{-3}{\sqrt{2}} & \frac{3}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}}_{X^T[\cdot][\cdot]} = \underbrace{\begin{pmatrix} 9 & -3 & 21 & -3 \\ 4 & 1 & 1 & 1 \\ 4 & -3 & 8 & 0 \\ 1 & 0 & 5 & -1 \end{pmatrix}}_{Y[\cdot][\cdot]}.$$

If we rearrange  $Y[\cdot][\cdot]$  as follows,

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$$\underbrace{\begin{pmatrix} \text{even rows and even columns} & | & \text{even rows and odd columns} \\ \text{odd rows and even columns} & | & \text{odd rows and odd columns} \end{pmatrix}}_{\text{rearranged } Y[\cdot][\cdot]},$$

we get  $\begin{pmatrix} 9 & 21 & -3 & -3 \\ 4 & 8 & -3 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & 5 & 0 & -1 \end{pmatrix}$ , that is just the 2D-DWT.

is implicitly we get  $\rightarrow Y = m(m \cdot s^+)^T$   
and rearrange  $Y$  as above

**Two dimensional Fourier transform**

► Observably,  $m[\cdot][\cdot]$  was first multiplied by the transpose of  $s[\cdot][\cdot]$ , i.e.,  $s^T[\cdot][\cdot]$ , as follows:

$$\underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}}_{m[\cdot][\cdot]} \cdot \underbrace{\begin{pmatrix} 4 & 2 & 1 & 0 \\ 6 & 6 & 4 & 3 \\ 10 & 8 & 6 & 2 \\ 12 & 12 & 7 & 1 \end{pmatrix}}_{s^T[\cdot][\cdot]} = \underbrace{\begin{pmatrix} \frac{10}{\sqrt{2}} & \frac{8}{\sqrt{2}} & \frac{5}{\sqrt{2}} & \frac{3}{\sqrt{2}} \\ \frac{-2}{\sqrt{2}} & \frac{-4}{\sqrt{2}} & \frac{-3}{\sqrt{2}} & \frac{-3}{\sqrt{2}} \\ \frac{22}{\sqrt{2}} & \frac{20}{\sqrt{2}} & \frac{13}{\sqrt{2}} & \frac{3}{\sqrt{2}} \\ \frac{-2}{\sqrt{2}} & \frac{-4}{\sqrt{2}} & \frac{-1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}}_{X[\cdot][\cdot]}$$

Now, multiplying  $m[\cdot][\cdot]$  by the transpose of  $X[\cdot][\cdot]$ , we have:

$$\underbrace{\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}}_{m[\cdot][\cdot]} \cdot \underbrace{\begin{pmatrix} \frac{10}{\sqrt{2}} & \frac{-2}{\sqrt{2}} & \frac{22}{\sqrt{2}} & \frac{-2}{\sqrt{2}} \\ \frac{8}{\sqrt{2}} & \frac{-4}{\sqrt{2}} & \frac{20}{\sqrt{2}} & \frac{-4}{\sqrt{2}} \\ \frac{5}{\sqrt{2}} & \frac{-3}{\sqrt{2}} & \frac{13}{\sqrt{2}} & \frac{-1}{\sqrt{2}} \\ \frac{3}{\sqrt{2}} & \frac{-3}{\sqrt{2}} & \frac{3}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}}_{X^T[\cdot][\cdot]} = \underbrace{\begin{pmatrix} 9 & -3 & 21 & -3 \\ 1 & 1 & 1 & 1 \\ 4 & -3 & 8 & 0 \\ 1 & 0 & 5 & -1 \end{pmatrix}}_{Y[\cdot][\cdot]}$$

If we rearrange  $Y[\cdot][\cdot]$  as follows,

$m$  is the filter for collecting data.  
notation

$$(ab)^T = b^T a^T$$

since the transpose can be distributed

$$Y = m(s \cdot m^+)$$

then rearrange

the inverse is pretty much:  $s = (m^+ Y) / m$

but first we rearrange again to get the original  $Y[\cdot][\cdot]$

# --- Two-dimensional Wavelet Transform ---

ORIGINAL IMAGE

 $s[.][.]$  $a[.][.]$  $b[.][.]$ 1<sup>st</sup> level  
2D-DWT $c[.][.]$  $d[.][.]$ 

$b[E-]E-]$  and  
 $c[E-]E-]$  are

for border, in  
they represent  
 abrupt changes

and  $d[E-]E-]$

is noise

Notably,  $b[.]$  and  $c[.]$  are more likely to be used for border extraction with emphasis on horizontal and vertical details, respectively, whereas  $d[.]$  shows high-frequency noise. In addition, we clearly observe that  $a[.][.]$  is practically identical to the original image but with a reduced size.

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2D-DWT is not used because  
it would be required the reconstruction  
of  $b, c, d$  which is not easy to interpret

$$2D-DWT_1 = 2D-DWT,$$

for part of the natural frequency etc  
in 2D is not applied

The image is deep but is the same  
but with small motion

the image is the same but for the  
whole motion

There's no probk with  
padding and border